

GarudaESE / EV60Y51A — Hardware Designer Confirmation Sheet

Purpose: Everything below was read from the EV60Y51A-V1.0 schematic by the firmware side. Please **confirm or correct each row** before we freeze port_config and the ADC map. Pins drive PPS, ADC channel allocation, and the ATA6847 bring-up, so a wrong pin = a non-booting build.

Legend (Source column): ✓ read directly off the MCU symbol (high confidence) · Δ from checklist/earlier read (verify) · ? not yet readable — please fill in.

Device on silk reads "MC505"; device-ID/pin-count indicate dsPIC33AK256MC506-E/M7, 64-pin VQFN. Q0. Confirm exact MCU orderable part number (...MC505 vs ...MC506, temp grade, package). □

A. Connectors — verified pinout

J1 — combined Programming / Debug / Control (BM12B-GHS-TBT, 13-pos incl. pin 0)

J1 pin	net	dsPIC pin	schematic pin-function string	Source	Confirm
0	NC (marked X)	—	—	✓	□
1	GND	—	—	✓	□
2	DEBUG_RX	RD8 (63)	CVDTX18/RP57/ASDA2/IOMAF3/RD8	✓	□
3	DEBUG_TX	RD7 (62)	CVDTX17/RP56/ASCL2/IOMAF4/RD7	✓	□
4	GND	—	—	✓	□
5	TELE_RX	RD4 (61)	CVDTX16/RP53/IOMBF6/RD4	✓	□
6	TELE_TX	RD6 (60)	RP55/PWM5L/ASDA1/IOMBD0/RD6	✓	□
7	DShot	RD5 (59)	RP54/PWM5H/ASCL1/IOMBD1/RD5	✓	□
8	PGC	RC2 (41)	PGC3/RP35/PWM4H/IOMAD7/RC2	✓	□
9	PGD	RC5 (42)	PGD3/RP38/PWM4L/IOMAD6/RC5	✓	□
10	GND	—	—	✓	□
11	+3.3 VDC	—	—	✓	□
12	MCLR	MCLR	(PWRGD-gated: R16 0R + R17 470R + C38)	✓	□

J2 — CAN (SM04B-GHS)

J2 pin	net	Confirm
1	GND	□
2	CAN_L	□
3	CAN_H	□
4	+5 V	□

Q-A1 (the big one — speed pot + arm). No ADC pin is on J1/J2 (all digital). BUT re-verification found a **reserved analog net Speed** on MCU pin 16 (RA6 / AD3AN2). - (a) ★ **Where does the Speed net go?** Is it routed to a connector / pad / test-point (i.e. usable for an external pot), or fed by an on-board divider, or is it an **unrouted reserved stub**? This is the single most useful answer for our bench control. □ - (b) If **Speed** is not usable, can a future rev route an ADC pin to J1 — e.g. the free no-connects **RB3 (pin 27, AD4AN3)** or **RB6 (pin 29, AD4AN1)**? □ - (c) **Arm switch:** OK to use a J1 digital pin as a GPIO input (internal pull-up, switch → GND)? Preferred — **DShot RD5**, or a DEBUG/TELE pin? Any external pull-ups already fitted? □

B. Motor PWM → ATA6847 (fixed-function pins)

net	dsPIC pin	schematic pin-function string	Source	Confirm
PWM1H	RD2 (51)	(PG1H)	Δ	□
PWM1L	RD3 (52)	(PG1L)	Δ	□
PWM2H	RD0 (49)	RP49/PWM2H/IOMAD3/RD0	✓	□
PWM2L	RD1 (50)	(PG2L)	Δ	□
PWM3H	RC3 (43)	RP36/PWM3H/IOMAD5/RC3	✓	□
PWM3L	RC4 (44)	RP37/PWM3L/IOMAD4/RC4	✓	□

Q-B1 (resolved — sanity-check). PWM → phase mapping traced on the bridge: **PWM1 → GH1/GL1 → Q1/Q4 → PHASE_U**, **PWM2 → PHASE_V**, **PWM3 → PHASE_W** (TP11/12/13). Confirm. □ **Q-B2.** Confirm **PWM1H/L = RD2/RD3**, **PWM2L = RD1** (only PWM2H/PWM3H/L were directly readable). □ **Q-B3.** Confirm ATA6847 input mapping & polarity: PWMxH → NBHx (**active-LOW** high-side), PWMxL → ILx (active-high). □ **Q-B4.** Confirm **dead-time is internal to the ATA6847** (so MCU dead-time ≈ 0); give the configured tCC / dead-time value. □

C. SPI2 ↔ ATA6847 — now READ (was blocker); please confirm

Resolved on re-verification: the bus is on the **SPI2-native pins**. | net | ATA6847 pin | dsPIC pin | port / function | Confirm | |---|---|---|---| | SPI_SCK | SCK 47 | 31 | RB10 = SCK2 (RP27) | ☐ | | SPI_SDI (ATA → MCU) | SDO 45 | 32 | RB11 = SDI2 (RP28) | ☐ | | SPI_SDO (MCU → ATA) | SDI 46 | 34 | RC9 = SDO2 (RP42) | ☐ | | SPI_NCS (GPIO) | NCS 44 | 33 | RC8 (RP41) | ☐ | | SPI_NIRQ (GPIO in) | NIRQ 41 | 35 | RC6 (RP39) | ☐ |

Q-C1. Confirm firmware should use **SPI2** (board routes SCK2/SDI2/SDO2), nCS/NIRQ as GPIO. ☐ **Q-C2.** Any SPI clock-rate constraint? (we plan BRG=14 → 3.33 MHz, ATA max 10 MHz). ☐

D. Comms / status (PPS / GPIO)

net	dsPIC pin	Source	Confirm
CAN_TX	RC10 (45)	△	☐
CAN_RX	RC11 (46)	△	☐
CAN_STBY	RB7 (30)	△	☐
Status LED (D4 orange)	RC0 (39)	△	☐

Q-D1. Confirm CAN_TX/RX/STBY pins and the transceiver (ATA6561). ☐ **Q-D2.** CAN bus termination R15 = 120 Ω fitted — is this board an **end node** or **mid-bus**? ☐ **Q-D3.** Confirm LED pin/colour (D4 orange = RC0); is the red LED (D5) populated? ☐

Q-D4 (GSP ↔ UART alignment — important for firmware). Our host protocol **GSP** (broker / MCP / GUI) talks over a UART, and it **must bind to the same physical UART/pins the board bring-up doc uses**. The board exposes two UARTs on J1: - **DEBUG** = DEBUG_TX **RD7 (62)** / DEBUG_RX **RD8 (63)** — the bring-up checklist references this one. - **TELEMETRY** = TELE_TX **RD6 (60)** / TELE_RX **RD4 (61)**.

Confirm: **GSP runs on the DEBUG UART (RD7/RD8)** — same pins the bring-up uses — and that DEBUG is a **full bidirectional** UART (not TX-only). State which **UART instance** (UART1/2/3) the bring-up firmware assigns to DEBUG so GSP uses the identical peripheral + PPS. If GSP should instead live on the TELEMETRY UART, say so. ☐

E. Analog / ADC map — needs full confirmation (critical for the 3-channel BEMF design)

Some analog nets were readable, others not. Please confirm **pin AND which ADC module (ADC1 / ADC2)** each lands on — we want the three BEMF phases on **independent channels** (no mux), so the ADC1/ADC2 split matters.

Re-verification read all of these directly. Please confirm **which ADC module (ADC1 / ADC2)** each ANx belongs to — we want the 3 BEMF phases on **independent channels** (no mux). | signal | net | dsPIC pin | port / ADC ANx | Confirm ADC# | |---|---|---|---| | Temperature (NTC) | TEMP | 1 | RA0 / AD3AN5 | ☐ ADC | | **Bus voltage** | **VBUS** | **2** | **RA7 / AD5AN0** | ☐ ADC | | BEMF_W (divider) | BEMF_W | 3 | RA1 / AD5AN1 | ☐ ADC | | **BEMFN (neutral)** | **BEMF_N** | **6** | **RA11 / AD5AN2** | ☐ ADC | | Op-amp ref (DAC) | UREF | 7 | RA8 / DACOUT2 | ☐ | | ATA_BEMF_U | ATA_BEMF_U | 8 | RA9 / AD1AN3 | ☐ ADC | | ATA_BEMF_V | ATA_BEMF_V | 9 | RA10 / AD1AN4 | ☐ ADC | | ATA_BEMF_W | ATA_BEMF_W | 24 | RB9 / AD2AN3 | ☐ ADC | | BEMF_U (divider) | BEMF_U | 17 | RB5 / AD3AN1 | ☐ ADC | | BEMF_V (divider) | BEMF_V | 23 | RB8 / AD2AN4 | ☐ ADC | | **Speed (reserved)** | Speed | **16** | RA6 / AD3AN2 | ☐ — see Q-A1(a) | | W-phase current (ATA CSA) | W_OA3_OUT | 15 | RA5 / AD3AN0 | ☐ ADC | | **DC-bus current (ATA CSA)** | **IBUSOA_OUT** | **28** | **RB4 / AD4AN0** | ☐ ADC | | U-phase current — OA1 OUT/IN-/IN+ | U_OA1 | 12/13/14 | RA2/RA3/RA4 | ☐ | | V-phase current — OA2 OUT/IN-/IN+ | V_OA2 | 20/21/22 | RB0/RB1/RB2 | ☐ | | (free, no-connect) | — | 27 / 29 | RB3/AD4AN3 - RB6/AD4AN1 | n/a |

Q-E1. Confirm the ADC1/ADC2 split for the BEMF phases — can BEMF_U/V/W (or ATA_BEMF_U/V/W) sample **simultaneously on independent channels**? ☐ **Q-E1b.** Confirm **Speed** (pin 16) is an intended analog throttle input and where it connects (ties to Q-A1a). ☐ **Q-E2.** Can the three phase BEMF inputs each be sampled on an **independent ADC channel simultaneously** (not muxed)? Confirm their ADC1/ADC2 distribution supports it. ☐

F. BEMF sensing architecture — RESOLVED on re-verification; please sanity-check

Two independent BEMF paths to the MCU, both wired: - (i) **BEMF_U/V/W + BEMF_N** = resistor-divider analog (24k/2k + 1000pF + BAT54; 3×24k star neutral) → MCU **RB5(17)/RB8(23)/RA1(3)/RA11(6)** ADC. - (ii) **ATA_BEMF_U/V/W** = ATA6847 **BEMF comparator DIGITAL outputs** (BEMF1/2/3, GDUCR1.BEMFEN=1) → MCU **RA9(8)/RA10(9)/RB9(24)** GPIO + TP8/7/6. **Q-F1.** Confirm both paths are populated and the pin assignment above. ☐ **Q-F2 (was open — now resolved).** Confirm **ATA_BEMF_U/V/W** are **digital comparator outputs** read as GPIO (matching the garuda-ak-ata-esc firmware), **not** analog. ☐ **Q-F3.** Virtual neutral **BEMF_N**: confirm 3×24k star → R68 680R + C63 6800pF. Was the heavier neutral filter (τ≈4.26 μs vs phase τ≈1.85 μs) intentional? ☐ **Q-F4.** Per-phase BEMF divider: confirm V & W mirror U (R61 24k / R69 2k / C64 1000pF / D13 BAT54) and the 0.0769 ratio / 42 V max. ☐

G. Current sense & shunts

signal	sense element	amp	nominal	Confirm
U phase	shunt (2 mΩ?)	dsPIC OA1 (≈×36.4)	73 mV/A	☐
V phase	shunt (2 mΩ?)	dsPIC OA2 (≈×36.4)	73 mV/A	☐
W phase	shunt (2 mΩ?)	ATA6847 CSA (×16)	32 mV/A	☐
DC bus	shunt (2 mΩ?, SL → GND)	ATA6847 CSA (×16)	32 mV/A	☐

Q-G1. Confirm all four shunt values (2 mΩ each?). ☐ **Q-G2.** Confirm which **ATA6847 CSA channel = W** vs = **DC-bus** (OPA2 vs OPA3), the CSA gain (×16), offset (−VBO2 ≈ 1.65 V), and **polarity** (inverted vs the dsPIC op-amps?). ☐ **Q-G3.** DC-bus shunt location — low-side bus return (SL → GND)? Sign convention for motoring current? ☐

H. Gate driver / power / clock / misc

Q-H1. Gate driver labelled **ATA6847T-5033** (note references a "5033L" variant). Confirm exact orderable suffix (the "L" affects VDH/VDDIO OV-shutdown bits). ☐ **Q-H2 (resolved — confirm).** Verified: **NRES pulled to VVIO via R3 10k** (not MCU-driven); **LH/RXD/TXD/LIN/WAKE = no-connect** → ATA configured **purely over SPI**. Confirm no MCU reset line is expected. ☐ **Q-H3.** Is **NIRQ** expected to be serviced (interrupt) or polled? Is the ATA watchdog required to be kicked in normal run? ☐ **Q-H4.** Charge pump: confirm the ATA uses its internal charge pump for the high-side (VG/VCP/VDH) — i.e., **no external bootstrap-cap pre-charge step** needed at start? ☐ **Q-H5 (resolved — confirm values).** MOSFET = **BSC028N06NS** (schematic suffix **NSSC** — confirm NSSC vs NSSG), 60 V ~2.8 mΩ. Gate net (verified): series **22 Ω** (R18/19/20), turn-off **10 Ω + PMEG3020EJ** (R24-26/D6-8), gate cap **1000 pF** (C42-44), pulldown **120 kΩ** (R21-23), source/LS series **3.3 Ω** (R27-32). Confirm. ☐ **Q-H6.** Bus: 6S, 25 V nominal — confirm **absolute-max bus voltage** and that VBUS divider (24k/2k, 0.0769) + D12 BAT54 clamp suit it. Input bulk C1/C2 = 330 μF/100 V, TVS D1 = SMAJ26A — confirm. ☐ **Q-H7.** Clock: Y1 = **8 MHz MEMS (DSA6001JA3B)** on OSC1/CLKI (pin 40) — confirm, and the intended **FCY / PLL** (system + peripheral clock targets). ☐ **Q-H8.** NTC: TH1 = 10 kΩ — provide **part / β value** (and R65 = 4.7 kΩ) for the temperature curve. ☐ **Q-H9.** Power rails: MCP16367 → 5 V, MCP1755 → 3.3 V, internal buck → VDDCORE ~1.1 V (L2). Confirm PWRGD → MCLR gating does **not** block ICSP, and the power-up sequencing. ☐ **Q-H10.** Spare PWM pins **PWM4H/L = RC2/RC5 (=PGC/PGD)** and **PWM5H/L = RD5/RD6 (=DShot/TELE_TX)** are shared with connector functions — confirm they are **not** independently used for braking / sync-rect, i.e. free for our use. ☐

I. Test points

Q-I1. List populated test points (TP1/2 = bus?, TP3 = 5 V, TP4 = 3.3 V, TP6/TP7 = ATA_BEMF?) and whether any expose an ADC-usable node we could borrow for the bench pot. ☐

J. Passive values — please confirm every value/designator

These were read from the schematic (0402 1% unless noted). They set ADC scaling, filter bandwidths, and ZC timing in firmware, so each needs confirmation. Where a designator is uncertain it is marked **(desig?)**.

J.1 BEMF sense — per phase (divider + filter + clamp)

node	top R	bottom R	filter C	clamp diode	ratio / τ	Confirm
BEMF_U	R61 = 24 k	R69 = 2 k	C64 = 1000 pF/50 V	D13 = BAT54XV2T1G → 3.3 V	0.0769 / τ≈1.85 μs	☐
BEMF_V	R?? = 24 k (desig?)	R?? = 2 k	C?? = 1000 pF	D14 = BAT54	same	☐
BEMF_W	R?? = 24 k (desig?)	R?? = 2 k	C?? = 1000 pF	D15 = BAT54	same	☐

Q-J1. Confirm V/W mirror U exactly (values + that each phase has its own 1000 pF + BAT54). Confirm the **42 V** full-scale assumption behind 0.0769. ☐

J.2 Virtual neutral (BEMF_N)

element	value	Confirm
Star resistors (PHASE_U/V/W → node)	R60 / R64 / R67 = 24 k each	☐
Series R to node	R68 = 680 Ω	☐
Filter C to GND	C63 = 6800 pF/50 V	☐

Q-J2. Confirm; note τ_N ≈ 4.26 μs (≈2.3× the per-phase 1.85 μs). Intentional? ☐

J.3 Phase current sense — dsPIC op-amps (U = OA1, V = OA2)

element (U / OA1)	value	mirror (V / OA2)	Confirm
Series in (IN+ / IN-)	R43 / R51 = 330 Ω	R40 / R42 = 330 Ω (desig?)	☐
To op-amp	R41 / R49 = 470 Ω	R44 / R45 = 470 Ω (desig?)	☐
Feedback R C	R54 = 12 k C54 = 22 pF	R52 = 12 k C55 = 22 pF (desig?)	☐
UREF bias R	R39 = 12 k	(mirror)	☐
Filter caps	C48 = 1500 pF (IN+ → GND), C51 = 2200 pF (diff) , C56 = 1500 pF (IN- → GND)	C49 / C52 / C57 = 1500/2200/1500 pF	☐
Gain	R54/R43 = 36.4 → 73 mV/A @ 2 mΩ	same	☐

Q-J3. Confirm U-chain values and that V mirrors it (V designators are inferred). ☐

J.4 W-phase & DC-bus current — ATA6847 CSA path

element	value	Confirm
Shunts (W, DC-bus)	2 mΩ each	☐
CSA input anti-alias	R56 / R58 = 4.7 Ω + C59 / C60 / C61 = 0.022 μF	☐
DC-bus CSA output RC	R5 = 360 Ω + C17 = 680 pF → IBUS_OA_OUT (fc≈650 kHz)	☐
W CSA output RC	R6 = 360 Ω + C18 = 680 pF → W_OA_OUT	☐
CSA gain / offset	×16 / offset ≈ VBO2 (~1.65 V)	☐

Q-J4. Confirm shunt values and the 360 Ω/680 pF output filters; confirm which CSA (R5/C17 vs R6/C18) is DC-bus vs W. ☐

J.5 VBUS & TEMP

node	elements	Confirm
VBUS	R59 = 24 k / R66 = 2 k (0.0769), C62 = 1000 pF/50 V , D12 = BAT54XV → 3.3 V	☐
TEMP	TH1 = 10 k NTC ($\beta = ?$), R65 = 4.7 kΩ (3.3 V → TH1 → TEMP → R65 → GND)	☐

Q-J5. Confirm VBUS divider/clamp and the NTC part + β value (needed for the temp curve). ☐

J.6 ATA6847 supply & charge-pump passives

node	value	Confirm
VS (pin 3)	FB1 = 600 Ω + C4 = 10 μF/50 V + C5 = 0.1 μF	☐
VDD1 (pin 48, 5 V LDO)	C7 = 2.2 μF/16 V	☐
VDD2 / VVIO (pin 1, 3.3 V LDO)	C8 = 2.2 μF/16 V	☐
VDH (pin 15)	C3 = 0.1 μF + C6 = 1 μF	☐
CPP2/CPN2 (17/16)	C9 = 0.22 μF/50 V (0603)	☐
VG (pin 9)	C10 = 3.3 μF/25 V (0805)	☐
CPP1/CPN1 (13/14)	C12 = 0.22 μF/50 V (0603)	☐
LH (pin 2)	R3 = 10 kΩ → VVIO	☐

Q-J6. Confirm charge-pump + supply decoupling values. ☐

J.7 Power input & reset

node	value	Confirm
Bus bulk	C1 / C2 = 330 μF/100 V	☐
Bus TVS	D1 = SMAJ26A	☐
AVDD filter	FB3 = 120 Ω + C29 1 μ F + C30 0.1 μ F	☐
VDDCORE (pin 56)	C34 = 10 μ F + C33 0.1 μ F (internal buck, L2)	☐
MCLR / PWRGD gate	R14, R16 = 0 Ω , R17 = 470 Ω , C38	☐

Q-J7. Confirm bulk/TVS, AVDD ferrite, VDDCORE buck inductor L2 value, and the MCLR gating network. ☐

J.8 Gate drive

element	value	Confirm
Gate series R (HS/LS)	R?? = (value?)	☐
Gate turn-off diodes	D6–D11 = PMEG3020EJ	☐
Gate-source pulldown (if any)	(value?)	☐

Q-J8 (resolved — confirm). Gate net read off the bridge: series **R18/19/20 = 22 Ω** , turn-off **R24-26 = 10 Ω** + **D6-8 = PMEG3020EJ**, gate cap **C42-44 = 1000 pF/50 V**, pulldown **R21-23 = 120 k Ω** , source/LS series **R27-32 = 3.3 Ω** , per-leg bus decoupling **C39-41 = 10 μ F/75 V**. Confirm. ☐

Reply inline in the Confirm column (✓ / correction). Anything in ? / (desig?) / (value?) rows is something we need from you. Pair with `GarudaESE_PinMap.md`.